GLOBAL AI HACKATHON · BIFROST + TAVILY CHALLENGES

AgentPulse

Which AI agent can you trust, and is its answer still true today?

Benchmarks agents through

Bifrost

Verifies answers live with

Tavily

agentpulse-9551.onrender.com

Picking an AI agent today is guesswork

Demos, not data

Teams choose a model or agent after trying a handful of prompts by hand.

Failures stay hidden

One run looks fine. Repeat it, or let a tool time out, and the answers change.

Stale answers sound confident

Models answer from old training data, with no hint that the world has moved on.

An agent, asked in November 2026:

> “The current UK Prime Minister is Rishi Sunak.”

Outdated. Andy Burnham has held the office since July 2026, and nothing in the answer says it might be wrong.

Two questions, one platform

01 · BENCHMARK THROUGH BIFROST

Which agent can I rely on?

Every agent gets the **same tasks**, several times over, with tool failures injected on purpose. It is scored on accuracy, consistency, recovery, speed and cost, and ranked by one **reliability score**.

02 · VERIFY WITH TAVILY

Is the answer still true today?

The answer is split into claims, and the **time-sensitive** ones are checked against live web results **with their publication dates**. Each claim comes back current, outdated or wrong, with what is true now.

How it works, in six steps

1 · MODELS

Pick models

Built-in Groq, OpenRouter and local Ollama, or your own endpoint and key.

2 · AGENTS

Build agents

A model plus instructions, tools (web search, calculator, date) and a step budget.

3 · SUITES

Choose tasks

Questions with their correct answer and a grading rule; built-in or your own.

4 · RUN BENCHMARK

Run it

Every agent × every task × repeats, with optional tool failures. Live progress.

5 · COMPARE · TRACES · REPORTS

Read the results

Who won and why, step-by-step replays, and Markdown, CSV and JSON exports.

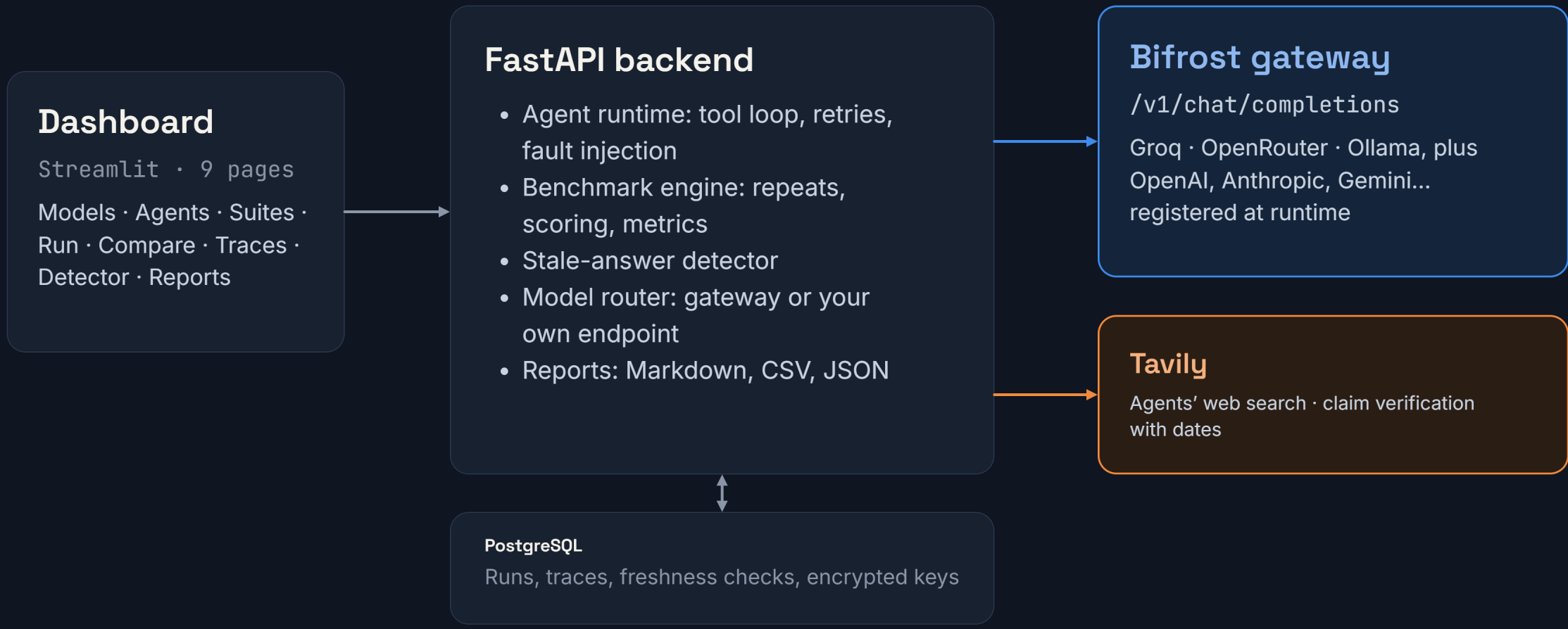
6 · STALE-ANSWER DETECTOR

Check freshness

Paste any answer, or let runs check theirs, against today's web.

Mock mode runs everything offline for free, on the same code paths; **live mode** calls real models and real search.

Architecture



HOW WE USE BIFROST · 1

One gateway for every model call

- **Same path, fair comparison.** Agents, the LLM judge and the stale-answer detector all call one OpenAI-compatible endpoint.
- **Any provider, one format.** A model is just a provider/model id: cloud, free tier or a local Ollama model.
- **Keys stay in the gateway.** Provider keys live in Bifrost's config; our backend never sees them.
- **Real cost.** Prices come from Bifrost's model catalogue, giving cost per run and cost per correct answer.
- **Errors become metrics.** Rate limits, provider errors and rejected tool calls drive retries, and feed the recovery score.

```
POST /v1/chat/completions

groq/openai/gpt-oss-20b

groq/qwen/qwen3.8-27b

openrouter/nvidia/nemotron-3.5-
lightning:free

ollama/granite4:3b (on a laptop)

gemini/gemini-flash-lite-latest (added
at runtime)
```

HOW WE USE BIFROST · 2

Your hosted model, registered inside Bifrost at runtime

Save a key
OpenAI, Anthropic, Gemini… on the Models page, encrypted at rest

→

Register
Management API adds the provider and key, with a setup token

→

Validate
Bifrost checks the key; the result is shown on the card

→

Self-heal
Bifrost restarted and forgot it? Re-register, retry once

It stays direct, and says why, when:

- it’s a self-hosted URL (Ollama, LM Studio, vLLM…)
- Bifrost already has a key for that provider: no mixing keys
- the gateway can’t accept it (no token or config store)

LIVE · REAL GEMINI KEY

“pong” in 2.5 s

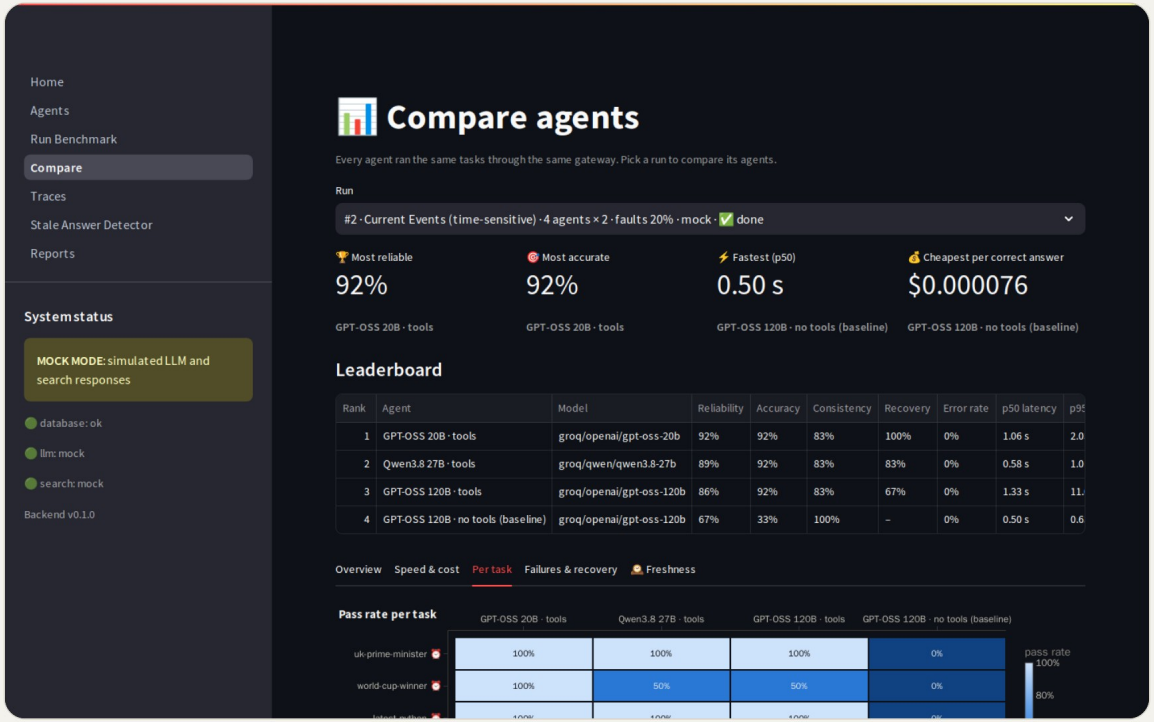
through Bifrost, with tool calling ✓, and 61 models discovered

HOW WE USE TAVILY · 1

Live web search inside the agents

- **A real tool.** Agents can call web_search, backed by Tavily: ranked sources, a summary and publication dates.
- **Tools vs no tools, measured.** The same model, with and without search, on identical ⏰ time-sensitive tasks.
- **Search can fail too.** We inject timeouts, empty and malformed results into the tool to see who recovers.

In the built-in demo run, the no-tools baseline fails exactly the ⏰ rows: it answers from training data.



HOW WE USE TAVILY · 2

The stale-answer detector

1 · BIFROST LLM

Extract claims

Split the answer into self-contained facts

2 · LLM + RULES

Time-sensitive?

Timeless facts are not searched, so no credits are spent on them

3 · TAVILY

Search today

A neutral query, dated sources, newest first

4 · BIFROST LLM

Judge

Current, outdated, contradicted or unverifiable

5 · SCORE

Freshness

Current ÷ decisive claims, plus what is true now

Neutral queries: “current UK prime minister”, never the claimed answer, so old articles that repeat it don’t confirm it.

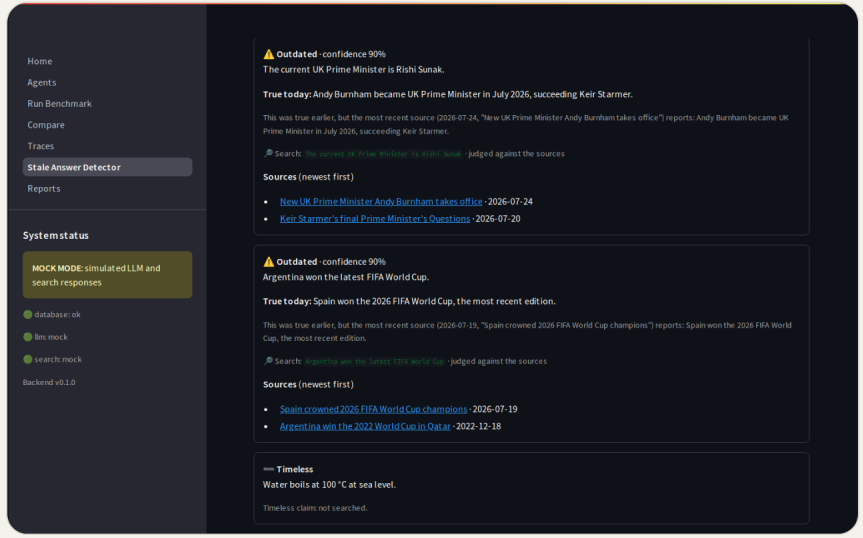
Automatic in runs: every answer to a ⏰ task is checked, and identical claims are searched once per run.

Outdated ≠ wrong: outdated means it was true once; contradicted means it never was.

Never crashes: a search or model failure becomes *unverifiable*, with the reason, and models fall back on rate limits.

Verified live against today's web

Claim in an answer	Verdict	Evidence (Tavily, dated)
The current UK PM is Rishi Sunak	Outdated	NPR, Georgetown, The Hill, Jul–Nov 2026: Andy Burnham
Argentina won the latest World Cup	Outdated	NBC, CNBC, Jul 2026: Spain won 2026
The newest Python is 3.13	Outdated	Python Insider, Oct 2026: 3.15
Elon Musk is the UK PM	Contradicted	Never true: Andy Burnham
Water boils at 100 °C at sea level	Timeless	Not searched: no credits spent
It is 2024	Outdated	Checked against the clock



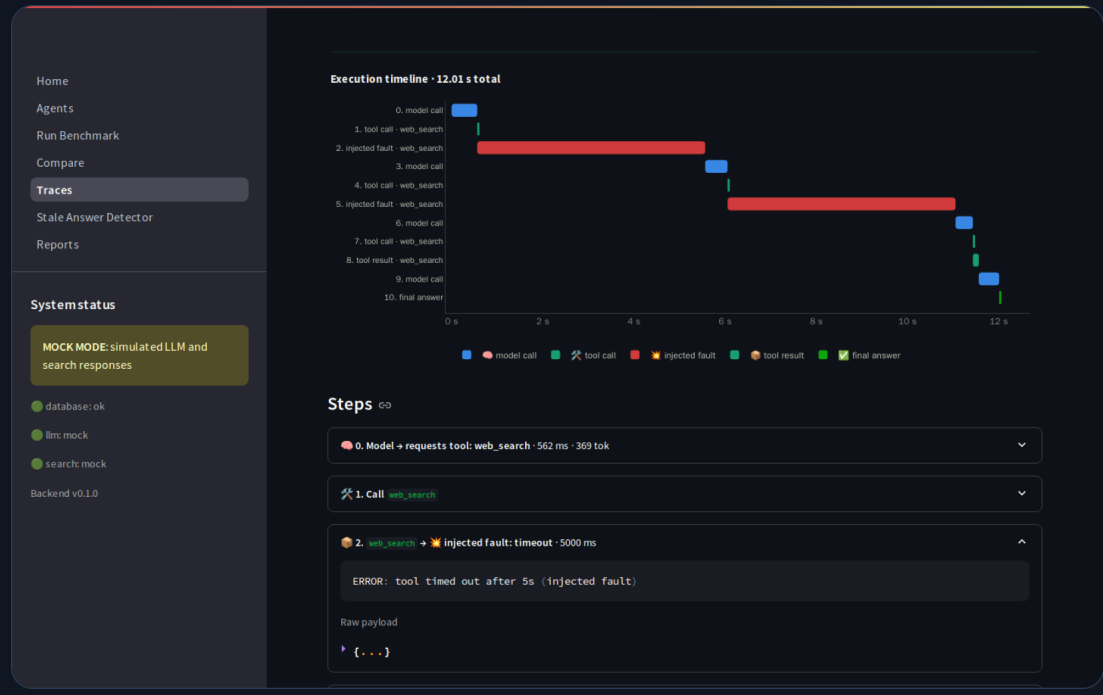
Agents are tested under failure

A real tool loop: the model calls web search, a calculator or today's date, reads the result and answers, within a step budget.

Injected faults: timeouts, HTTP 500s, empty and malformed tool results, at a rate you choose, and reproducible.

Real-world retries: backoff on transient errors, longer waits on rate limits, and a corrective retry when a provider rejects a tool call.

Every step traced: each model call, tool call, fault and retry, with tokens and time, replayable on the Traces page.



A reliability score you can explain

40%

Accuracy: answers graded correct

25%

Consistency: same result each repeat

20%

Recovery: right despite failures

15%

Stability: no errors

Parts that don't apply (consistency with one repeat, recovery when nothing failed) are dropped and the weights renormalised, so the score always stays 0–100%.

pass@k / pass-all-k: can it, and does it always?

Cost per correct answer

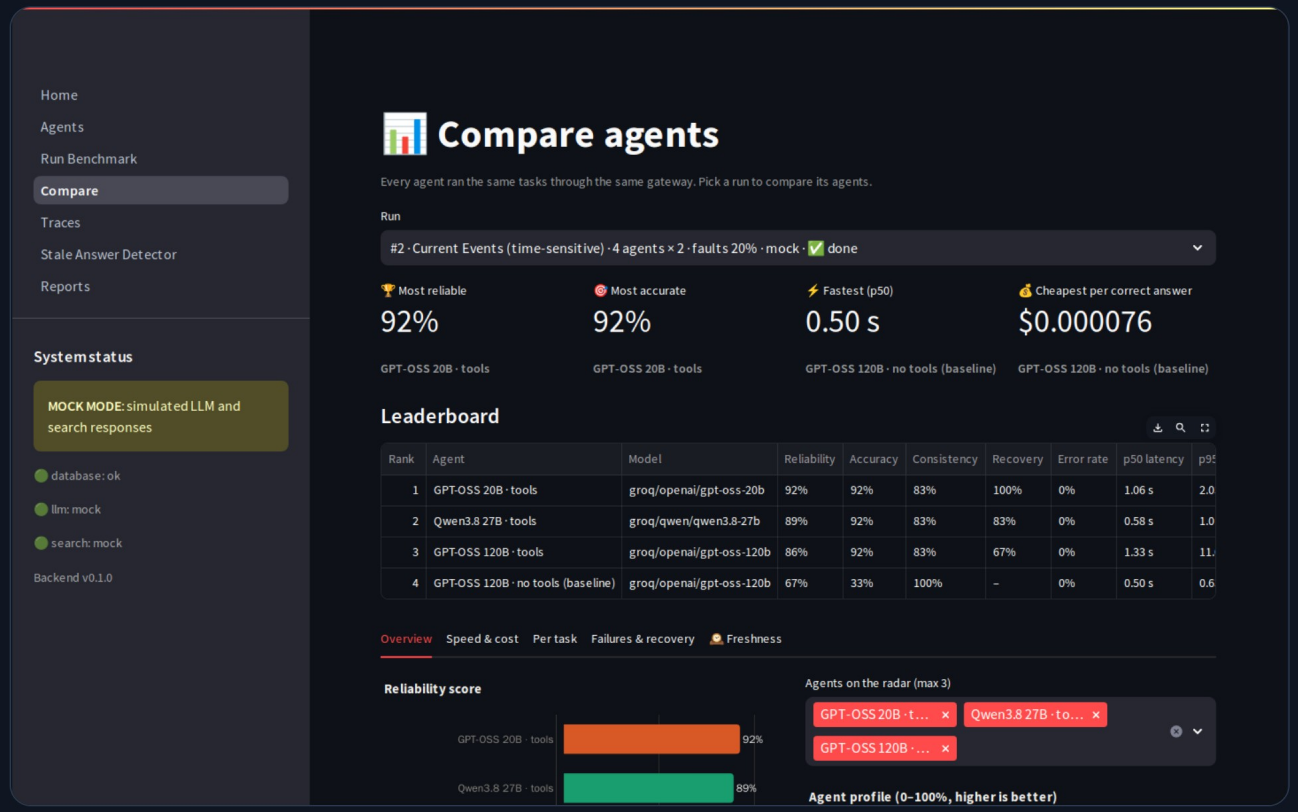
p50 / p95 latency: typical and worst-case speed

Steps and tool calls per task

Tokens and cost: list price via Bifrost

Freshness: shown beside the score

The Compare dashboard



Headline tiles: most reliable, most accurate, fastest, cheapest per correct answer

Overview: reliability bars and a radar of three agents

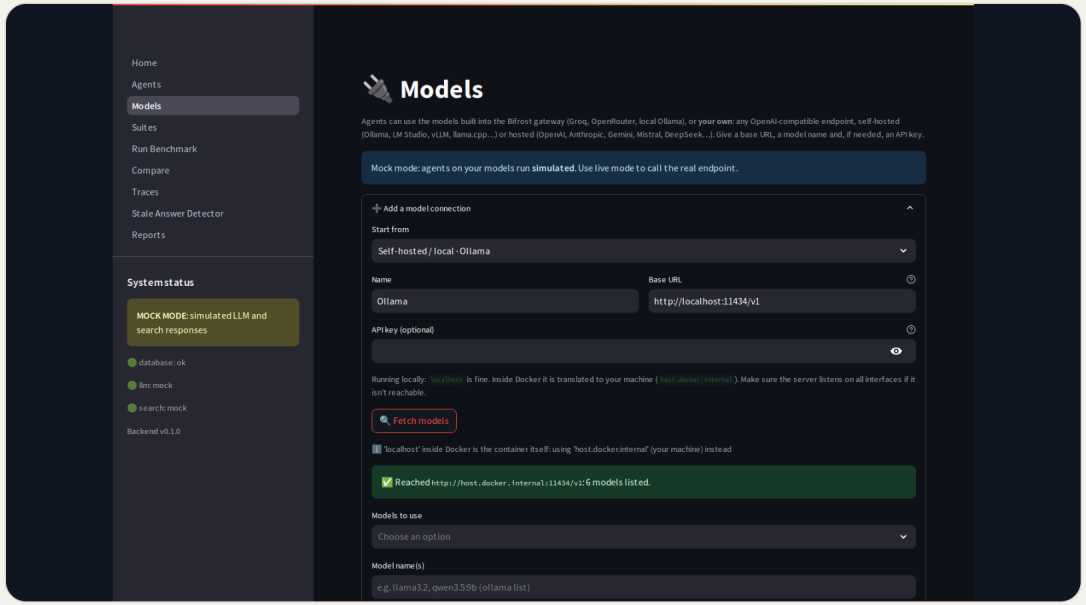
Speed and cost: latency, accuracy vs cost, tokens

Per task: a pass-rate heatmap

Failures and recovery, with a trace link on every failure

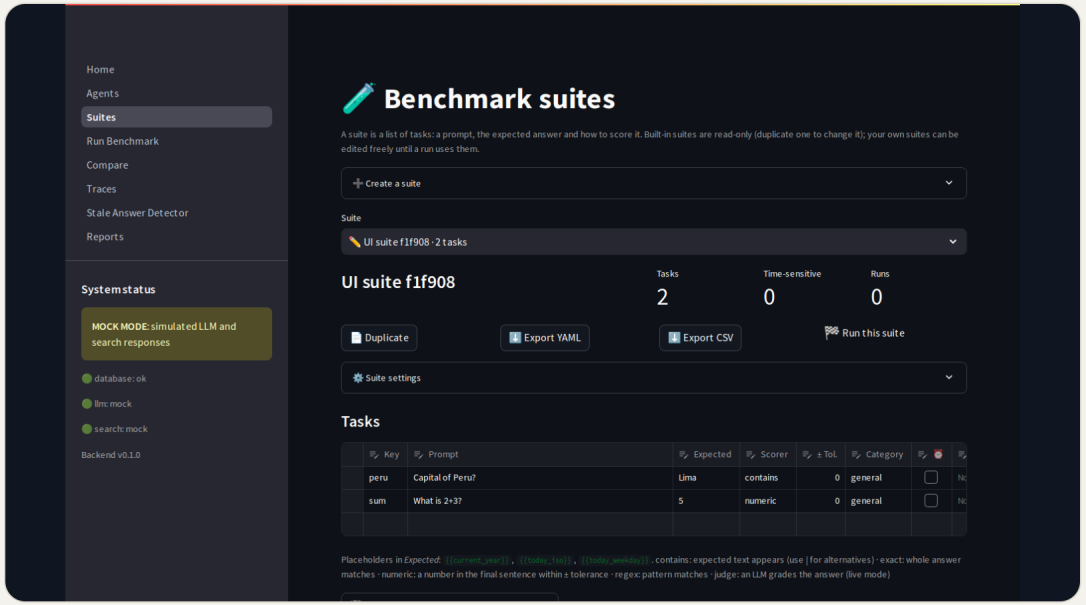
Freshness: whose answers were out of date, and why

Bring your own models and benchmarks



🔌 Models: any OpenAI-compatible endpoint

Ollama, LM Studio, vLLM, llama.cpp or a hosted API: base URL + model + optional key. Keys are encrypted and never shown again.



🧪 Suites: your own questions

Edit tasks in a table, import or export CSV/YAML, and test a grading rule on a sample answer before you run anything.

It works with real models

7/7

live checks pass through Bifrost: Groq, OpenRouter, a local Ollama model, tool calling and Tavily search

2.5 s

a Gemini key registered in Bifrost at runtime: first answer through the gateway, tool calling ✓

100% · $0

a 3B laptop model via its own URL matched Groq's GPT-OSS 20B on Math & Reasoning, at 7.6 s vs 1.0 s typical

3.14

what both live agents called the latest Python; flagged stale with Tavily, since 3.15 shipped in October 2026

AgentPulse · 15 · live runs, 5–6 Nov 2026

Built to be trusted, too

Mock mode, same code

Simulated models and search follow the exact same pipelines, so the demo works offline and for free, with example data on first open.

400 automated tests

318 backend, 24 unit and 58 dashboard tests, plus a 17-check smoke test of the deploy image. CI runs on every push.

Secure by default

Encrypted API keys, never returned; keys and account ids scrubbed from errors; private addresses blocked on the public demo.

Fails honestly

Rate limits fall back to other models, Bifrost registrations self-heal, and every failure says why, instead of crashing.

Stack: **FastAPI · Streamlit · PostgreSQL · Plotly · Bifrost · Tavily · Docker Compose · GitHub Actions · Render**

Live, deployed, and open to try

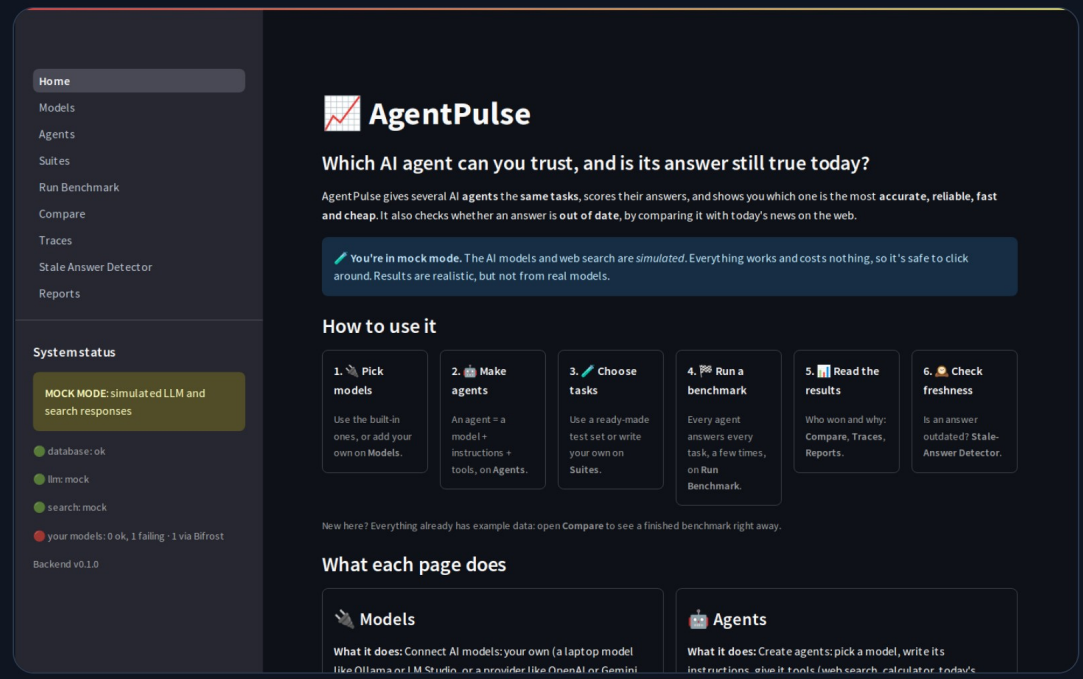
Public demo on Render: everyone gets mock mode, which is free and can’t drain anyone’s credits.

An access code unlocks live mode: real models through Bifrost and real Tavily search.

Ships itself: every push runs CI, and Render redeploys only when it passes.

Runs anywhere: one command starts Bifrost, the API, the dashboard and Postgres in Docker.

```
make setup && make up
```



AGENTPULSE

One gateway for every model. Live truth for every answer.

Bring your models and your questions. See which agent you can trust, and whether its answer is still right today.

Try it

`agentpulse-9551.onrender.com`

Code

`github.com/UTSAVS26/AgentPulse`